Quadratics Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Quadratic formula: the −b

ORIGINAL PROBLEM

Solve $x^2 - 6x + 5 = 0$ with the quadratic formula.

INCORRECT WORKED SOLUTION

$a = 1, b = -6, c = 5$

$$x = \frac{-6 \pm \sqrt{36 - 20}}{2}$$

$$x = \frac{-6 \pm 4}{2}$$

$x = -1$ or $x = -5$

A. EXPLAIN THE MISTAKE (1 POINT)

The formula starts with $-b$. Since $b = -6$, $-b = +6$, not -6.

B. SHOW CORRECT WORK (1 POINT)

$a = 1, b = -6, c = 5$

$$x = \frac{6 \pm \sqrt{36 - 20}}{2}$$

$$x = \frac{6 \pm 4}{2}$$

$x = 5$ or $x = 1$

2. Substituting c

ORIGINAL PROBLEM

How many real solutions does $2x^2 + 3x - 2 = 0$ have? Solve it.

INCORRECT WORKED SOLUTION

$a = 2, b = 3, c = 2$

$b^2 - 4ac = 9 - 4(2)(2) = -7$

The discriminant is negative: no real solutions.

A. EXPLAIN THE MISTAKE (1 POINT)

The constant is $c = -2$, not 2. The discriminant is $9 - 4(2)(-2) = 25$, so there are two real solutions: $x = \frac{-3 \pm 5}{4}$, so $x = \frac{1}{2}$ or $x = -2$.

B. SHOW CORRECT WORK (1 POINT)

$a = 2, b = 3, c = -2$

$b^2 - 4ac = 9 - 4(2)(-2) = 25$

$$x = \frac{-3 \pm 5}{4}$$

$x = \frac{1}{2}$ or $x = -2$

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3. Square root method

ORIGINAL PROBLEM

Solve $(x - 2)^2 = 25$

INCORRECT WORKED SOLUTION

$(x - 2)^2 = 25$

$x - 2 = 5$

$x = 7$

A. EXPLAIN THE MISTAKE (1 POINT)

Taking a square root gives TWO answers: $x - 2 = 5$ or $x - 2 = -5$. So $x = 7$ or $x = -3$.

B. SHOW CORRECT WORK (1 POINT)

$(x - 2)^2 = 25$

$x - 2 = \pm 5$

$x = 2 + 5 = 7$ or $x = 2 - 5 = -3$

4. Completing the square

ORIGINAL PROBLEM

Solve $x^2 + 6x = 7$ by completing the square.

INCORRECT WORKED SOLUTION

$x^2 + 6x + 36 = 7 + 36$

$(x + 3)^2 = 43$

$x = -3 \pm \sqrt{43}$

A. EXPLAIN THE MISTAKE (1 POINT)

Add (half of b)$^2 = 3^2 = 9$ to both sides, not $6^2 = 36$. Then $(x + 3)^2 = 16$, so $x + 3 = \pm 4$, and $x = 1$ or $x = -7$.

B. SHOW CORRECT WORK (1 POINT)

$x^2 + 6x + 9 = 7 + 9$

$(x + 3)^2 = 16$

$x + 3 = \pm 4$

$x = 1$ or $x = -7$

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5. Finding the vertex

ORIGINAL PROBLEM

Find the vertex of $y = x^2 - 4x + 1$.

INCORRECT WORKED SOLUTION

$a = 1, b = -4$

$x = \frac{b}{2a} = \frac{-4}{2} = -2$

$y = (-2)^2 - 4(-2) + 1 = 13$

Vertex: $(-2, 13)$

A. EXPLAIN THE MISTAKE (1 POINT)

The axis of symmetry is $x = \frac{-b}{2a}$, with a NEGATIVE b. $x = \frac{4}{2} = 2$, $y = 4 - 8 + 1 = -3$, so the vertex is $(2, -3)$.

B. SHOW CORRECT WORK (1 POINT)

$a = 1, b = -4$

$x = \frac{-b}{2a} = \frac{4}{2} = 2$

$y = 2^2 - 4(2) + 1 = -3$

Vertex: $(2, -3)$

6. Reading vertex form

ORIGINAL PROBLEM

Identify the vertex of $y = (x + 3)^2 - 2$.

INCORRECT WORKED SOLUTION

$y = (x + 3)^2 - 2$

$h = 3, k = -2$

Vertex: $(3, -2)$

A. EXPLAIN THE MISTAKE (1 POINT)

Vertex form is $y = (x - h)^2 + k$. Writing $x + 3$ as $x - (-3)$ shows $h = -3$. The vertex is $(-3, -2)$.

B. SHOW CORRECT WORK (1 POINT)

$y = (x - (-3))^2 + (-2)$

$h = -3, k = -2$

Vertex: $(-3, -2)$

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7. Opening direction

ORIGINAL PROBLEM

Find the vertex and tell whether it is a maximum or minimum for $y = -2x^2 + 8x$.

INCORRECT WORKED SOLUTION

$a = -2$

The parabola opens up.

The vertex is a minimum.

A. EXPLAIN THE MISTAKE (1 POINT)

The sign of a decides the direction. $a = -2$ is NEGATIVE, so the parabola opens DOWN and the vertex is a maximum. $x = \frac{-8}{2(-2)} = 2$, $y = 8$, maximum at $(2, 8)$.

B. SHOW CORRECT WORK (1 POINT)

$a = -2$ (negative), so it opens down

$x = \frac{-8}{2(-2)} = 2$

$y = -2(2)^2 + 8(2) = 8$

Maximum value 8 at (2, 8)

8. Interpreting solutions

ORIGINAL PROBLEM

A ball's height is $h = -16t^2 + 64t$. When does it hit the ground again?

INCORRECT WORKED SOLUTION

Set $h = 0$: $-16t^2 + 64t = 0$

$-16t(t - 4) = 0$

$t = 0$

The ball lands at 0 seconds.

A. EXPLAIN THE MISTAKE (1 POINT)

There are TWO solutions: $t = 0$ (when the ball is launched) and $t = 4$ (when it lands). The ball lands after 4 seconds.

B. SHOW CORRECT WORK (1 POINT)

$-16t(t - 4) = 0$

$t = 0$ or $t = 4$

$t = 0$ is the launch.

The ball lands after 4 seconds.

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9. Simplifying the formula

ORIGINAL PROBLEM

Simplify $x = \frac{-2 \pm \sqrt{12}}{2}$

INCORRECT WORKED SOLUTION

$x = \frac{-2 \pm \sqrt{12}}{2}$

$x = -1 \pm \sqrt{12}$

A. EXPLAIN THE MISTAKE (1 POINT)

The 2 in the denominator must divide BOTH terms in the numerator, and $\sqrt{12} = 2\sqrt{3}$ first. $x = \frac{-2 \pm 2\sqrt{3}}{2} = -1 \pm \sqrt{3}$.

B. SHOW CORRECT WORK (1 POINT)

$\sqrt{12} = 2\sqrt{3}$

$x = \frac{-2 \pm 2\sqrt{3}}{2}$

$x = -1 \pm \sqrt{3}$

10. Isolating x^2

ORIGINAL PROBLEM

Solve $2x^2 = 8$

INCORRECT WORKED SOLUTION

$2x^2 = 8$

$x^2 = 16$

$x = \pm 4$

A. EXPLAIN THE MISTAKE (1 POINT)

Divide both sides by 2: $x^2 = \frac{8}{2} = 4$, not 16. Then $x = \pm 2$.

B. SHOW CORRECT WORK (1 POINT)

$2x^2 = 8$

$x^2 = \frac{8}{2} = 4$

$x = \pm 2$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points